

Laserify: From Multimodal Input to Fabrication-Ready Laser Cut Plans

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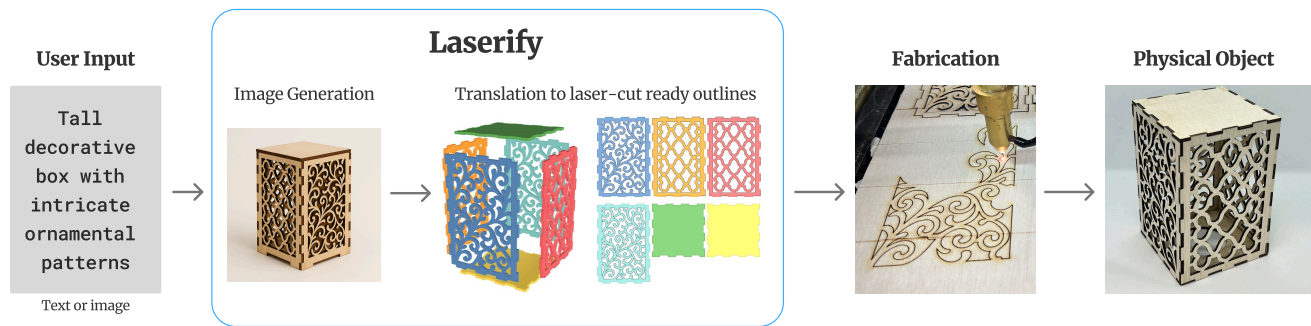


Figure 1: Proposed hybrid pipeline: from multimodal image generation to fabricated laser-cut model

Abstract

Laser cutting enables rapid fabrication of functional objects, yet translating visual ideas into manufacturable cut files remains a challenge for non-experts. We present Laserify, a hybrid pipeline that transforms 2D renderings of laser-cut objects into fabrication-ready 2D cut plans. Our pipeline supports diverse starting points—text descriptions, sketches, or inspiration photos—by leveraging existing generative AI tools to produce reference images as input. Our approach decomposes this translation into two phases: semantic interpretation, where vision-language models identify parts, infer spatial relationships, and generate outlines; and algorithmic geometric processing, where deterministic methods enforce fabrication constraints. This division leverages AI for ambiguous interpretive tasks while guaranteeing physical viability through geometric validation. Preliminary comparison with direct VLM-to-CAD baselines suggests that the staged approach yields more fabrication-viable outputs, particularly for objects with complex curved outlines. By reducing CAD literacy while maintaining manufacturing precision, this work lowers the bar for translating creative ideas into physical artifacts.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Accessibility design and evaluation methods; • **Computing methodologies** → Scene understanding; • **Applied computing** → Computer-aided design.

Keywords

AI-assisted design, democratization, digital fabrication, fabrication workflows, laser cutting, multimodal AI, vision-language models

ACM Reference Format:

Ziv Botzer, Guy Austern, and Yoav Sterman. 2026. Laserify: From Multimodal Input to Fabrication-Ready Laser Cut Plans. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798604>

1 Introduction

The accessibility of consumer-grade fabrication tools has ushered in an era of “everyday making,” empowering non-experts to repair, customize, and create objects in their homes or in neighborhood fab-labs [18, 22]. Among these increasingly available technologies, laser cutting is particularly valued for its operational efficiency, allowing for the rapid fabrication of useful, medium-to-large objects [2]. Simultaneously, advances in generative AI have transformed how design ideas can be expressed and explored. Text-to-image generation, sketch interpretation, and multimodal AI tools now enable rapid visual ideation, allowing users to quickly generate and iterate on design concepts. However, a significant disparity persists between this ease of visual ideation and the technical rigor required for physical production. As noted by Carrington et al.



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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798604>

[4], even when users possess a clear visual goal, they struggle with the “technical burden” of translating conceptual ideas into manufacturable CAD¹ files.

While Vision-Language Models (VLMs)—large language models (LLMs) extended with image understanding capabilities—can interpret visual inputs, they fundamentally lack the precise spatial reasoning required for manufacturing. Recent studies indicate that end-to-end VLM generation of 3D geometries is prone to hallucinations, disconnected topology, and invalid constraints [11, 14]. Conversely, traditional parametric tools ensure geometric validity but lack the semantic intuition to interpret a user’s rough sketch or photograph [9].

To bridge this gap, we present **Laserify**, a hybrid pipeline that transforms multimodal inputs into fabrication-ready laser-cut plans. Drawing on neurosymbolic design principles [20], our approach decomposes the workflow into three stages: (1) *reference image generation*, where diverse inputs—text descriptions, sketches, or reference photos—are translated into realistic images of laser-cut objects; (2) a *semantic interpretation* stage, where VLMs identify parts, infer spatial relationships, and generate part outlines; and (3) an *algorithmic processing* stage, where deterministic algorithms enforce fabrication constraints such as kerf compensation, joinery, and material thickness. This division of labor leverages the interpretive flexibility of AI while guaranteeing the physical viability of the final artifact.

Modern generative AI can translate abstract representations into concrete visual references, enabling diverse starting points for creating new objects—text descriptions, hand sketches, or photos of existing artifacts. We focus on images as the entry point to the core pipeline because they offer a productive balance: users can evaluate and refine a generated image before committing to fabrication, images are easily manipulated through photo editing and text-based AI tools, and image generation is now fast enough to support rapid iteration. Moreover, images convey geometric detail beyond what text alone can express [11].

Contributions. This work contributes: (1) a multi-stage pipeline architecture that decomposes multimodal input to fabrication translation into reference image generation, VLM-driven semantic interpretation, and algorithmic processing; (2) a progressive constraint strategy where each stage narrows the solution space and produces verifiable intermediate representations; and (3) a comparison against single-step baselines: direct VLM-to-outlines and VLM-to-OpenSCAD² generation.

2 Related Work

Our approach draws from several complementary research areas: automated 3D reconstruction from visual inputs, CAD reverse engineering for fabrication, the emerging application of VLMs to design workflows, and interactive laser-cutting systems.

3D reconstruction from visual inputs: Recent advances in single-image and text-to-3D reconstruction have made generating 3D geometry increasingly accessible. Large Reconstruction Models such as TripoSR [23] and InstantMesh [26] can produce visually compelling meshes from a single photograph, while text-to-3D

systems [7, 13, 16, 17] enable generation directly from natural language prompts. However, these approaches prioritize visual fidelity over manufacturability—outputs are topologically arbitrary meshes lacking semantic part-segmentation and the planar constraints required for laser cutting. Even more recently, methods such as CADDreamer [12] and Img2CAD [27] attempted to close this gap by generating structured CAD representations directly from single images — producing B-reps and parametric models rather than raw meshes. However, these approaches target volumetric 3D solids and mechanical part geometries, leaving the translation to planar fabrication assemblies unaddressed. Reconstructing parametric CAD models from visual inputs requires resolving not only geometry but also assembly logic and fabrication constraints. As Kyaw et al. [10] argue, even structured generative outputs neglect physical constraints such as connectivity and joinery. The assembly logic that distinguishes a fabrication-ready design for planar laser-cut assemblies, from a generic, geometrically plausible model is not addressed in these approaches.

LLMs and VLMs in design: VLMs are increasingly being applied to CAD generation, with examples ranging from scripts for the construction of simple primitives [24] to recent multimodal approaches which map images directly to parametric formats. Wu et al. [25] demonstrate that combining sketch images with textual representations improves CAD auto-completion accuracy, suggesting that visual and linguistic modalities provide complementary geometric information. However, direct LLM generation remains brittle: LLMs excel at discrete semantic reasoning — identifying parts, selecting operations, decomposing structure — but struggle to satisfy precise geometric constraints, producing parts that float, intersect, or misalign [14, 27], with performance degrading further as geometric complexity increases [6, 11]. This brittleness extends to fabrication contexts, where globally applied AI edits risk breaking physical functionality [5]. These findings motivate our hybrid approach: leveraging VLMs for interpretive tasks while delegating fabrication constraints to algorithmic methods.

Laser Cutting Workflows: Prior work on laser-cutting interfaces has primarily focused on assisting manual modeling or on converting 3D inputs into 2D plans. Tools like Kyub [2] enable constrained voxel-based modeling, while Platener [3], LaserOrigami [19], and Roadkill [1] automate conversion of 3D meshes to planar fabrication files—the latter optimizing layouts for fast assembly. However, these workflows assume precise 3D inputs—unavailable for original creations by novice makers—or require manual modeling expertise. Sketch-based interfaces like SketchChair [21] and FlatFitFab [15] lower barriers by inferring structure from drawings, yet still require users to explicitly define geometry.

Our work extends this trajectory: rather than manual sketching, we use generative visual interpretation to translate ambiguous images—such as photographs or AI-generated references, now accessible to anyone, into fabrication-valid assemblies, reducing technical barriers for everyday makers.

¹Computer Aided Design

²OpenSCAD (<https://openscad.org>) is a scripting language for parametric CAD.

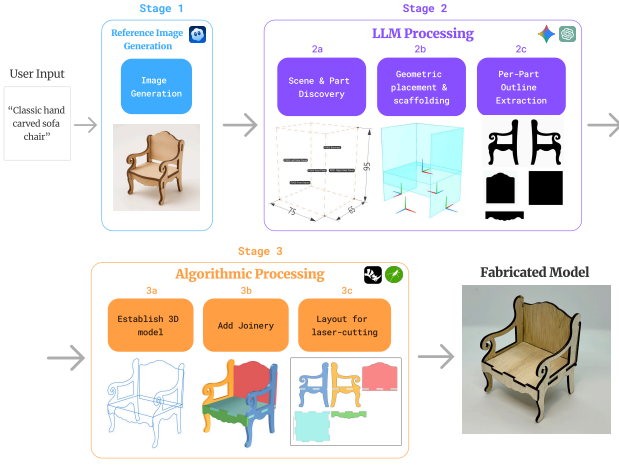


Figure 2: The Laserify pipeline: multimodal inputs are translated into images of laser-cut objects (Stage 1), processed through VLM-driven semantic interpretation (Stages 2a–2c), and refined via algorithmic processing (Stages 3a–3c) to produce fabrication-ready cut plans and physical artifacts.

3 Methods

The Laserify pipeline comprises three stages (Figure 2): image generation, VLM³-driven semantic interpretation, and algorithmic fabrication processing. Rather than asking an VLM to generate complete CAD code in a single step—an approach prone to geometric inconsistencies—we decompose the problem into constrained subtasks with verifiable intermediate outputs [8].

3.1 Stage 1: Multimodal Input to Image

Stage 1 builds upon the success of current image generation tools, and translates diverse user inputs—text descriptions, hand sketches, or reference photographs—into realistic images of laser-cut objects. Normalizing varied starting points into a consistent visual representation allows users to evaluate and refine the concept before committing to fabrication. (Figure 3).

3.2 Stage 2: VLM-Driven Semantic Interpretation

Stage 2a (Scene and Part Discovery) establishes global context: the model estimates material properties and scale, outputting a structured JSON containing a coordinate frame, the part inventory, and spatial relationships. **Stage 2b (Geometric Placement and Scaffolding)** defines construction planes and 2D bounding rectangles for each part, validating the geometric relationships against attachment constraints. **Stage 2c (Per-Part Outline Extraction)** generates part outlines through AI image synthesis rather than direct code generation: a generative model creates binary masks of individual parts at viewing angles derived from Stage 2b, which are traced into vectors using Potrace⁴. While direct SVG generation

³We use **VLM** to denote models that jointly process images and text, sometimes referred to as *multimodal large language models (MLLMs)* in recent work.

⁴<https://potrace.sourceforge.net>

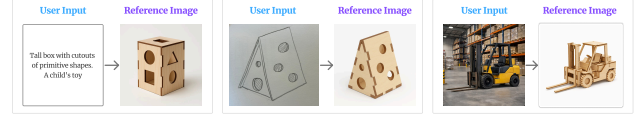


Figure 3: Diverse inputs: text descriptions, hand sketches, or reference objects—are translated in Stage 1 into realistic images of laser-cut models, providing a unified starting point for subsequent pipeline stages.

using VLMs often produces self-intersecting paths or non-closed loops and struggles with intricate outlines, binary masks can capture fine geometric detail at full image resolution. These masks can easily be converted into topologically sound vectors ready for algorithmic processing by applications such as Potrace. We use Gemini (gemini-3-pro-preview) for Stages 2a–2b (selected for semantic reasoning) and GPT (gpt-5.1-2025-11-13) for Stage 2c (selected for image generation accuracy).

3.3 Stage 3: Algorithmic Fabrication Processing

Stage 3 applies deterministic geometric processing using Grasshopper, ensuring fabrication constraints are handled by proven algorithmic methods rather than by VLM spatial reasoning. Unlike VLMs, which struggle to satisfy multiple simultaneous constraints and are prone to hallucination under geometric complexity, algorithmic methods reliably handle intricate geometries while preserving precision. This stage is fully automated by default, requiring no parametric modeling expertise. However, users may optionally adjust parameters such as model scale or material thickness, with joinery automatically regenerated to maintain fabrication validity. **Stage 3a** reconstructs the 3D assembly by positioning 2D outlines in their computed construction planes and extruding to material thickness. **Stage 3b** generates joinery (tabs, slots, finger joints) at part intersections with kerf compensation. **Stage 3c** arranges parts on cutting sheets using nesting algorithms and exports fabrication-ready DXF files.

4 Results

We compared Laserify against two baseline approaches: (1) **Direct outline extraction**, where a VLM is prompted to generate ready-to-cut part outlines in a single step—producing a 2D fabrication plan showing all parts required to recreate the model. GPT (gpt-5.1-2025-11-13) was selected for its superior image generation accuracy. (2) **OpenSCAD generation**, where a VLM is prompted to generate executable CAD code from the input image. Here, Gemini (gemini-3-pro-preview) was selected for its stronger performance in semantic reasoning and code generation.

We assessed outputs qualitatively across three dimensions: semantic correspondence (correct part count and inter-part relationships), geometric fidelity (outlines closely resembling those visible in the input image), and fabrication viability (can the output actually be cut and assembled?). Figure 4 shows representative results across four test objects.

Direct outline extraction failed most severely: outputs exhibited inconsistent scaling, hallucinated parts, and while joinery was

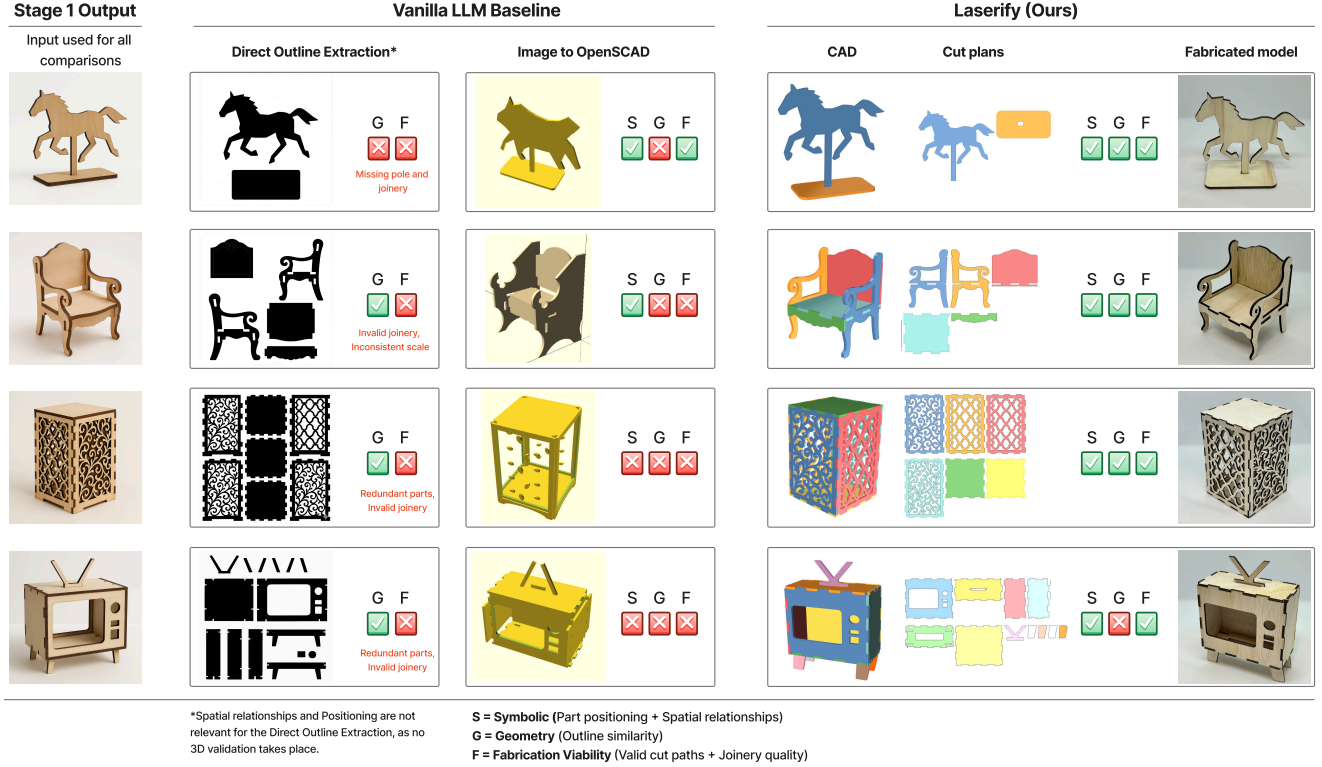


Figure 4: Comparative results across four test objects showing Laserify’s advantage in producing fabrication-viable outputs

sometimes generated, parts had no coherent relationship—joints did not align or fit together. Results were unusable for fabrication.

The **OpenSCAD generation** produced reasonable part decompositions but struggled with geometric precision—complex curves often resulted in illegal geometries and non-manufacturable outputs, and spatial positioning frequently failed with floating or intersecting parts. This confirms that direct VLM-to-CAD generation breaks down when translating intricate visual details into executable code.

Laserify produced fabrication-viable outputs across test objects, as can be seen in the fabricated images in Figure 4, successfully handling complex curved outlines that would challenge manual CAD modeling. The staged approach enabled the system to leverage VLM world knowledge for completing occluded parts while relying on algorithmic stages to enforce fabrication constraints. Limitations emerged with complex multi-part assemblies and non-orthogonal connections, where even structured prompting struggled to maintain geometric consistency.

5 Discussion and Conclusion

We presented Laserify, a hybrid pipeline that transforms multi-modal inputs into fabrication-ready cut plans of laser-cut models by decomposing the problem into VLM-driven semantic interpretation followed by algorithmic processing. The core contribution lies in the system architecture: successful AI-assisted fabrication

requires dividing labor between VLM-driven methods (scene interpretation, spatial inference) and deterministic algorithms (joinery, kerf, layout validity). This division leverages AI where ambiguity and semantic understanding are strengths, while relying on proven geometric methods where precision and constraint satisfaction are non-negotiable.

Preliminary evaluation suggests that VLMs excel at semantic reasoning—identifying parts, inferring spatial relationships, and completing occluded structures—but struggle with complex multi-part assemblies requiring precise geometric constraint satisfaction, particularly for non-orthogonal connections. The inherent stochasticity of VLM outputs complicates evaluation: variability can reflect legitimate interpretive ambiguity or lead to hallucinations. As foundation models evolve with improved spatial reasoning, future work will explore techniques such as providing VLMs with geometry evaluation tools and enabling iterative self-verification loops to improve geometric scaffolding accuracy.

By reducing CAD literacy requirements while maintaining fabrication precision, this work demonstrates how hybrid AI-algorithmic approaches can democratize design-to-fabrication workflows—offering a template for tools that balance creative flexibility with technical rigor, and a concrete step toward making digital fabrication accessible to everyday makers.

6 Supplementary Materials

Full prompts used in Stage 2 and baselines are available at <https://github.com/zivbot/Laserify>. The specific version of these materials used for this work is preserved under the Git tag v1.0.0-CHI2026.

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